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Concept-based heterogeneous context fusion for enhancing RAG via AMR-driven distillation☆

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ABSTRACT

Retrieval Augmented Generation (RAG) enhances Large Language Models (LLMs) on knowledge-intensive tasks by involving external documents to support their inference. However, the retrieved documents inherently form a heterogeneous context, because individual sources differ in relevance, noise level, and informational completeness, which makes the aggregated context internally inconsistent. Naively concatenating such contexts dilutes the model's attention and degrades inference accuracy. Motivated by the way humans selectively attend to essential concepts when processing complex information, we propose a concept-based heterogeneous context fusion framework driven by Abstract Meaning Representation (AMR) distillation. The framework uses AMR as a structured intermediate representation to parse each retrieved document into a semantic graph, from which an AMR-driven distillation algorithm extracts the informative concepts into a compact, unified set for downstream reasoning. A key advantage is that AMR's abstraction principle normalizes surface-level variations across sources, producing cross-document consistent concepts with reduced linguistic noise. Extensive experiments on the open-domain Q&A datasets PopQA and EntityQuestions show that the proposed method consistently outperforms existing baselines, with gains becoming more pronounced as the number of supporting documents grows, while maintaining robustness across diverse backbone LLMs.

1. Introduction

Large Language Models (LLMs) have emerged as indispensable tools for knowledge-intensive tasks, owing to their extensive parametric knowledge memorized through pre-training [1]. However, the presence of outdated or incorrect knowledge within their internal parameters can lead to hallucinations, hindering the accuracy of downstream inference [2]. This limitation is particularly pronounced when handling long-tail knowledge for domain-specific queries, as the inherent difficulty in memorizing rare entities persists even in the most robust models. Consequently, the overreliance on potentially flawed parametric knowledge significantly reduces the reliability of LLMs' outputs, especially in scenarios with fine-grained knowledge requirements [3,4].

Retrieval Augmented Generation (RAG) addresses this challenge by incorporating external documents to augment LLMs' inference with query-focused, non-parametric knowledge [5]. However, the retrieved documents collectively form a heterogeneous context that poses a critical challenge for fusion. On the one hand, multiple retrieved documents differ substantially in relevance, noise level, and informational completeness, making the aggregated context internally inconsistent [6]. On the other hand, the surface-level textual sequences alone cannot expose the underlying semantic structure needed to identify and consolidate the most informative concepts across documents. When such heterogeneous context is directly concatenated as LLM input, noisy and irrelevant information dilutes the model's attention and degrades inference performance [7]. Therefore, effectively distilling the most supportive knowledge for heterogeneous contextual information fusion remains a critical challenge for RAG [8].

Studies have shown that individuals tend to search for key concepts when reading long documents, as the brain completes the remaining details based on prior patterns, knowledge, and context [9]. This selective attention to critical information allows readers to bypass redundant

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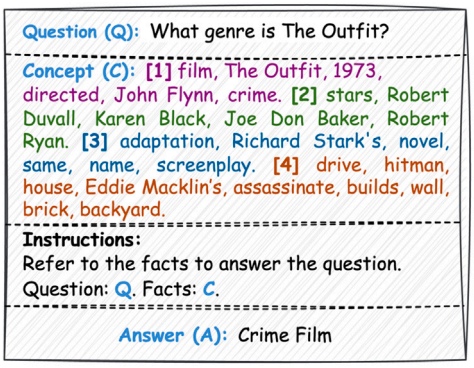


Fig. 1. An example of concept-based RAG. The concept block shows distilled concepts for each sentence. This setting enables the LLM to answer the question using only concepts rather than full sentences.[1]

details and reorganize the text around its core meaning [10]. As illustrated in Fig. 1, given only the key concepts from question-related supporting documents, one can still grasp the essential semantics. LLMs, having parameterized massive common knowledge, exhibit a similar ability in context understanding even when word- or character-level information is disrupted [11]. This cognitive insight suggests that fusing heterogeneous contexts does not require preserving all raw information, but rather distilling the essential concepts across documents. Linguistic features such as semantic and syntactic structures have been shown to improve the interpretability, controllability, and diversity of Natural Language Generation (NLG) [12,13]. Language models can implicitly discover these features in pre-training to maintain the coherence of the generated text [14]. Explicitly leveraging linguistic features for downstream tasks is also beneficial, as it restructures source documents into concise representations that reduce entropy by concentrating on the critical information, thereby supporting a more accurate understanding of the described scenarios [15]. This advantage makes linguistic features a promising tool that introduces semantically rich concepts to fuse key information in heterogeneous contexts.

Inspired by the aforementioned insights, we propose a concept-based heterogeneous context fusion framework that distils key concepts from retrieved documents to produce a unified, compact knowledge representation for enhancing RAG. To effectively extract informative concepts from heterogeneous contexts, we introduce Abstract Meaning Representation (AMR), a semantic formalism that encodes the meaning of serialized texts as a rooted, directed, labeled, acyclic graph [16]. Compared to other linguistic representations, AMR prioritizes semantic consistency among the concepts carried by its nodes, offering the advantage of automatically normalizing surface-level variations and resolving abbreviated terms, so that the fused concepts capture the underlying meaning beyond the limitations of linguistic noise [17]. In our framework, AMR serves as a foundation of concept distillation structure, where the proposed AMR-driven distillation algorithm parses raw documents into AMR graphs and formats the most informative concepts into a unified set for LLM inference. We evaluate the concept-based RAG framework on two open-domain Q&A datasets, PopQA [18] and EntityQuestions [19]. The results show that our method improves significantly as the number of supporting documents increases, outperforming baselines with various compression methods and backbone LLMs. The contributions of this paper are as follows:

- We propose a concept-based heterogeneous context fusion framework for RAG that integrates AMR-driven structured semantic distillation, enabling LLMs to focus on essential rather than noisy knowledge when processing retrieved supporting documents. To our knowledge, this is the first work to introduce AMR-driven distillation to fuse heterogeneous retrieval contexts for enhanced inference.
- We propose an AMR-driven concept distillation algorithm that serves as the core fusion mechanism. It transforms unstructured documents into structured AMR graphs and then distils informative concepts into a unified representation. The resulting concept set is knowledge-concentrated compared to the raw retrieved documents, reducing the interference of irrelevant information during inference.
- Extensive experiments on open-domain Q&A datasets demonstrate that our framework effectively enhances inference performance and outperforms baselines with various context compression and fusion methods across multiple backbone LLMs, confirming the applicability of concept-based heterogeneous context fusion in knowledge-intensive RAG scenarios.

The remainder of this paper is organized as follows: Section 2 reviews related works in heterogeneous data fusion, long-context understanding, and structure-augmented generation. Section 3 details the proposed concept-based RAG framework, including the AMR-driven distillation algorithm. Section 4 introduces the datasets, baselines, and evaluation metrics. Section 5 presents the results and analysis. Finally, Section 6 concludes the paper and presents directions for future work.

2. Related work

2.1. Heterogeneous context fusion and compression

Heterogeneous data fusion aims to integrate information from diverse sources or representations to achieve a more accurate and comprehensive understanding than any single source alone [20]. Within retrieval-augmented generation, heterogeneous fusion manifests at two levels: across multiple retrieved documents that differ in relevance, noise, and informational granularity, and across different representational forms of the same textual content [12]. The aggregated retrieval context in RAG systems is therefore heterogeneous both in source quality and in representational form. Some work has attempted to address the challenges posed by long and noisy contexts in LLMs [21]. Dai et al. [22] introduced segment-level recurrence in Transformer-XL to retain longer context, while Beltagy et al. [23] proposed sparse attention in Longformer to efficiently process longer sequences. However, such modifications require complex re-training. In contrast, prompt compression compresses long-token prompts into low-dimensional prompts [24], providing a lighter alternative but limiting transferability across LLMs.

Recent efforts focus on expanding the context window or using explicit compression. Chen et al. [25] used position interpolation to scale LLaMA's context to 32k tokens with minimal fine-tuning. Ding et al. [26] proposed LongRoPE, extending context to 2048k tokens using a progressive strategy while preserving short-context performance. However, longer contexts risk introducing redundant and heterogeneous information [27]. To mitigate this, Li et al. [28] filtered irrelevant content based on self-information, and Chuang et al. [29] developed Nano-Capsulator to reduce inference latency via prompt compression. These methods operate at the lexical level, which helps LLMs concentrate on key knowledge in the massive contexts.

[1] The corresponding complete sentences: **[1]** The Outfit is a 1973 crime film directed by John Flynn. **[2]** It stars Robert Duvall, Karen Black, Joe Don Baker and Robert Ryan. **[3]** Flynn's screenplay is an adaptation of the novel of the same name by Richard Stark. **[4]** Two hitmen drive to Eddie Macklin's house to assassinate him as he builds a brick wall in his backyard.

Table 1
Summary of the main notation used in this section.

Symbol	Description
$\mathcal{Q}$	Input question
$\mathcal{D} = \{D_1, \ldots, D_K\}$	Set of K retrieved supporting documents
$\mathcal{A} = \{\mathcal{A}_1, \ldots, \mathcal{A}_M\}$	Set of M candidate answers
$\mathcal{G} = \langle \mathcal{N}, \mathcal{E} \rangle$	AMR graph with node set $\mathcal{N}$ and edge set $\mathcal{E}$
$n \in \mathcal{N}$	A single node carrying a concept
$\mathcal{C}$	Distilled concept set
K, M	Number of documents and answers

2.2. Structure-augmented generation

Applying linguistic principles in LLMs improves coherence and semantic fidelity [30]. Augmentation with syntactic trees [31] and lexical patterns [13] injects linguistic features, aiding in faithful generation. Ahmed et al. [32] enhanced code summarization via semantic prompt augmentation, while Zhou et al. [33] introduced InstructCTG, a syntax-constrained generation framework allowing adaptability without modifying model architecture. These linguistic augmentation methods can be viewed as introducing a structured data modality that is heterogeneous to the raw text, providing complementary information for improved generation quality. AMR offers structured semantic representations for capturing core concepts [34], serving as a graph-based modality rather than sequential text. Frisoni et al. [35] integrated AMR to improve biomedical summarization, and Ribeiro et al. [36] used it to detect factual inconsistencies in summarization. Shi et al. [37] proposed AMR-TST for generating fluent, styled texts via core concept optimization. Jangra et al. [38] used AMR for style transfer while preserving semantic content. These works highlight AMR's strength in encoding informative linguistic features as a structured representation modality.

Another structure-augmented generation is knowledge graphs. Zhu et al. [39] proposed KG^{2}RAG, which retrieves seed chunks by semantic matching and then uses a knowledge graph for organizing. Hu et al. [40] introduced GRAG, which retrieves textual subgraphs and integrates their textual and topological information into generation. The methods in this line depend on externally curated knowledge graphs. In contrast, our approach derives concepts from the intrinsic semantics of the source, keeping the method self-contained and applicable to open-domain text that a curated knowledge graph may not cover. Integrating structured AMR for heterogeneous fusion and generation is a promising direction to reduce the reliance on external information sources.

3. Methods

3.1. Concept-based RAG framework

The proposed concept-based heterogeneous context fusion framework is illustrated in Fig. 2, and Table 1 summarizes the notation. Given an input question $\mathcal{Q}$, the (a) information retrieval component employs a retriever to return the top-K supporting documents $\mathcal{D} = \{D_1, \ldots, D_K\}$ relevant to $\mathcal{Q}$ from sources such as Wikipedia or other repositories. At this stage, the retriever's performance significantly influences the answer set $\mathcal{A} = \{\mathcal{A}_1, \ldots, \mathcal{A}_M\}$ [41]. We adopt two mainstream retrievers in RAG systems, Contriever [42] for the PopQA dataset and BM25 [43] for the EntityQuestions dataset. We hypothesize that all retrieved supporting documents $\mathcal{D}$ contain the correct answer corresponding to $\mathcal{Q}$, expressed as a proposition: $\forall D_k \in \mathcal{D}$, $\exists \mathcal{A}_m \in \mathcal{A}$, $\mathcal{A}_m \subseteq D_k$.

The (b) AMR-driven concept distillation component serves as the core heterogeneous context fusion mechanism. It leverages AMR as a structured intermediate tool to parse each retrieved supporting document D_k into an AMR graph $\mathcal{G}$, and then distils the most informative concepts $\mathcal{C}$ from these graphs into a compact concept set. This process operates on the semantic structure exposed by AMR. It consolidates heterogeneous multi-source contexts into a representation. The source contexts differ in relevance, noise, and redundancy, and the representation filters out concept-irrelevant information.

After obtaining the fused concept set $\mathcal{C}$, the (c) concept-based inference component integrates it with various backbone LLMs to derive answers $\mathcal{A}$ using the specific prompt template as follows: "[*Refer to the following facts to answer the question. Facts:* $\mathcal{C}$. *Question:* $\mathcal{Q}$]". The intensity of prompts has been demonstrated to influence LLMs' adherence to knowledge from internal memory and retrieved documents [44]. Since our assumption is that the retrieved documents contain correct answers, we encourage the LLMs to rely on the knowledge represented in $\mathcal{C}$ when responding to queries. This strategy helps minimize potential conflicts caused by their memorized parametric knowledge. To this end, we designate the concept set as a "*fact*" within the instructional prompt, explicitly creating a bounded scope for LLMs to treat the knowledge conveyed by $\mathcal{C}$ as correct. This non-parametric knowledge can integrate into LLMs in a plug-and-play manner. The overall framework can be represented as Eq. (1).

$$P(\mathcal{A}|\mathcal{Q}) = P(\mathcal{A}|\mathcal{C}, \mathcal{Q}) P(\mathcal{C}|\mathcal{D}, \mathcal{Q}) P(\mathcal{D}|\mathcal{Q}). \tag{1}$$

3.2. AMR-driven concept distillation

Abstract Meaning Representation (AMR) serves as a formal semantic structure that captures common-sense knowledge for representing events, temporal relations, participants, and other elements within serialized texts [45]. Given a supporting document $D_k \in \mathcal{D}$, the AMR parser converts D_k into the AMR graph $\mathcal{G} = \langle \mathcal{N}, \mathcal{E} \rangle$, where $\mathcal{N}$ represents the nodes carrying concepts and $\mathcal{E}$ denotes the edges that indicate correlations. The AMR graph exposes the underlying semantic structure of each source document, providing a reliable basis for identifying and consolidating the most informative concepts across multiple heterogeneous documents. Importantly, the abstraction principle of AMR maps each source document to a shared semantic representation. This mapping filters out surface-level variations among documents, such as differences in phrasing, abbreviations, and syntactic structure [17]. This normalization is particularly beneficial for multi-source heterogeneous context fusion, as documents retrieved from different sources often describe overlapping concepts using inconsistent surface forms. By distilling concepts from the normalized AMR structure rather than directly from the raw text, the resulting concept set achieves greater cross-document consistency and reduces the interference caused by surface-level noise. In this work, we employ a mBart-based [46] parser[2] trained on the AMR 3.0 corpus[3] to address potential multilingual concerns contained in the heterogeneous retrieved documents.

We propose the AMR-driven concept distillation algorithm to extract and format the informative concepts from $\mathcal{G}$, as described in Algorithm 1. This algorithm constitutes the core fusion mechanism that distils key concepts from the structured AMR graphs of multiple heterogeneous source documents into a unified concept set. The supporting document D_k contains multiple sentences (snt_n), and the AMR parser structurally parses D_k into a pre-defined `multi-sentence` structure. The SplitSnt($\cdot$) function partitions $\mathcal{G}$ and organizes the resulting sentence-based sub-graphs according to sequential order. We simplify $\mathcal{G}$ by disregarding the agent and patient of the concepts, i.e., the edges denoting relations between the connected concepts (Frame args, `ARGX`). As a result, $\mathcal{G}$ is streamlined into a unidirectional connecting structure. Based on this structure, we perform a Depth First Search, DFS($\cdot$), over the nodes $n \in \mathcal{N}$ of $\mathcal{G}$ to traverse the concepts. This approach preserves the connection

[2] https://github.com/BramVanroy/multilingual-text-to-amr.
[3] https://catalog.ldc.upenn.edu/LDC2020T02.

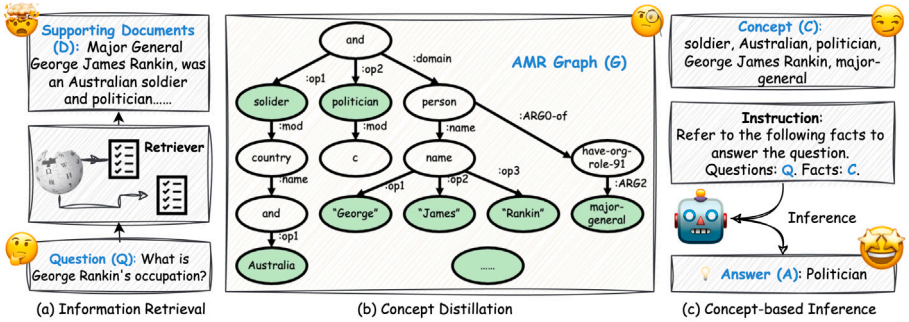


Fig. 2. The overview of the concept-based heterogeneous context fusion framework, which consists of three main components: (a) information retrieval, (b) AMR-driven concept distillation for heterogeneous context fusion, and (c) concept-based inference.

Algorithm 1: AMR-driven Concept Distillation

Input : AMR Graph ($\mathcal{G}$)
Output: concept (C)
Function Concept_Distillation($\mathcal{G}$):
  concept ← [], role ← [];
  for $\mathcal{G}_{\text{snt}_n}$ in SplitSnt ($\mathcal{G}$) **do**
    for n in DFS($\mathcal{G}_{\text{snt}_n}$) **do**
      if IsRole(n) **then**
        if IsName(n) **then**
          AppendRole(HandleName(n))
        if IsWiki(n) **then**
          AppendRole(HandleWiki(n))
        if IsDate(n) **then**
          AppendRole(HandleDate(n))
      else
        if role is not None **then**
          AppendConcept(HandleRole(role));
          role ← [];
        AppendConcept(n);
      if (n is Last) and (role is not None) **then**
        repeat : Algorithm.Line 5-11
        AppendConcept(HandleRole(role));
  concept ← ConceptFormat (concept);
  concept ← ConceptBacktrace (concept);
  return C ← concept

as it exists in the preceding sequential representation, and the process is illustrated in Fig. A.1. Previous research has investigated the influence of context order on LLMs [7]. We explore different traversal methods and evaluate their potential impact in Appendix C.

The AMR specification defines a set of roles to represent the semantic structure of sentences. In this paper, we focus on three roles: `:name`, `:wiki`, and `date-entity`, using IsRole(·) to identify them comprehensively. The `:name` role represents a property node within the AMR graph, denoting entities such as individuals, organizations, or geographic locations. When the concept expressed by `:name` spans multiple words, the AMR parsing process decomposes each word into predicate roles (`:op`), thereby dispersing the complete concept across multiple nodes. During the DFS(·) traversal, fragmented nodes can potentially confuse LLMs due to incomplete meaning expressions. To preserve the integrity of concepts carried by `:name`, we introduce HandleName(·), organizing predicates in a stack structure. The `:wiki` role provides reliable external concept references sourced from Wikipedia. To standardize the diverse surface-level expressions referring to the same named entities across different source documents, we use the HandleWiki(·) function, which aligns the concepts with the corresponding definitions in Wikipedia. If the concept in `:name` differs from `:wiki`, we designate the concept expressed by this node as `:wiki` to resolve semantic ambiguity. In addition, a `date-entity` role depicts temporal concepts. In our algorithm, we handle the roles `:year`, `:month`, and `:day` through HandleDate(·). This function consolidates roles under the same `date-entity`, forming concepts such as "19 04 2026" with numerical months translated into textual representations, "19 April 2026", for clear expression. AMR uses sense-specific annotations for certain nodes, such as `work-01`. The number indicates a specific sense of a word based on OntoNotes [47], a large-scale corpus with predicate–argument structures and sense distinctions for English verbs and nouns. For example, `work-01` refers to being employed, while `work-02` refers to functioning correctly. In our RAG scenario, we provide LLMs with supporting documents consisting of a fused concept set. This means that concepts are understood in relation to relevant contexts rather than in isolation. Therefore, the proposed framework relies on the contextual learning capability to distinguish between polysemous concepts, rather than relying on detailed semantic references. The nodes belonging to the aforementioned roles are integrated into the preliminary concept set through HandleRole(·), while AppendConcept(·) directly integrates the remaining nodes based on instances.

The AMR structure includes a collection of canonical nodes (`city-district`, `market-sector`, etc.) designed to enforce knowledge and prevent hallucination regarding entity types. However, in the concept-based fusion scenario, the inference process is not directly based on AMR but on the fused concepts. The auxiliary semantics embedded within these canonical nodes, which are absent in the source supporting documents, may dilute the core concept information. To address this concern, we employ ConceptFormat(·) to filter out these nodes and reduce potential interference. Furthermore, the selection of representations in AMR follows the principle of abstraction and generalization rather than exact lexical items. This representational choice may cause nodes to omit variations such as tense, which are informative for concept-based inference without reference annotations. To mitigate this, we develop the ConceptBacktrace(·) function to maintain consistency with the concepts expressed in the source supporting documents. This function facilitates the backtracing of formatted concepts by incorporating representations from the supporting documents, ensuring that they closely adhere to the original semantics without deviation. The backtraced concepts serve as the finalized fused concept set C, providing a unified representation that supports LLMs' inference. An example of the concept distilled by the proposed algorithm is in Table A.1.

4. Experiments

4.1. Datasets

We conducted extensive experiments to verify the effectiveness of the concept-based heterogeneous context fusion framework on open-domain Q&A datasets: PopQA [18] and EntityQuestions [19]. The

Table 2
Statistical results of the number of "<Q-A-D>" pairs from the datasets.

K =	1	2	3	4	5	6	7	8	9	10
PopQA [18]	738	1307	422	262	161	151	108	79	66	70
EntityQuestions [19]	1671	1127	670	454	335	264	196	166	163	103

experiments involved verifying the LLMs' inference performance with different K, which denotes the number of supporting documents provided for Q. Table 2 reports the statistics of selected pairs under different K settings.

4.2. Baselines

The baseline evaluations cover two aspects: (1) exploration of diverse backbone LLMs, and (2) experimentation with different context compression methods. Specifically, we adopt mainstream publicly available LLMs as backbones, including GPT-Neo-1.3B, GPT-Neo-2.7B [48], OPT-1.3b, OPT-2.7b [49], bloom-560m, bloom-1b1, bloom-1b7, bloom-3b [50], LLaMA-2-7b-chat-hf, LLaMA-2-13b-chat-hf [51]. The backbone LLMs coupled with the raw supporting documents serve as the Vanilla methods. Regarding the alternative aspect, we explore three context compression methods: context keywords extraction, context summarization, and Selective Context (SelCon) [28]. These methods aim to validate the effectiveness of context fusion and compression while preserving essential information for inference, covering discrete key features, fluent representation, and non-redundant information.

Inspired by Chuang et al. [29], we employ LLaMA-2-13b-chat-hf [51] for prompt-based baselines: context keyword extraction and summarization. This process involves extracting key phrases or terms from the context and generating a concise summary of the provided content, constrained by Prompt 1 and Prompt 2. For the SelCon baseline, we set the reduction ratio to 0.5 following a moderate compression setting. This setting retains approximately half of the tokens and provides a fair basis for comparison. These baseline settings demonstrate the advantages of the proposed AMR-driven distillation algorithm in capturing informative concepts when compared to various alternative fusion and compression techniques, whether prompt-based or semantic-based methods.

Prompt 1: Keywords Extraction

```
[INST] «SYS» Extract a few keywords from the following content. «/SYS»
Prompt = """Below is an instruction that describes a task, paired with an input that provides content.
### Instruction: {""" + Instruction + """}
### Input: {""" + D + """}
### Response: """ [/INST]
```

Prompt 2: Summary Generation

```
[INST] «SYS» Generate a short summary of the following content. «/SYS»
Prompt = """Below is an instruction that describes a task, paired with an input that provides content.
### Instruction: {""" + Instruction + """}
### Input: {""" + D + """}
### Response: """ [/INST]
```

4.3. Evaluation metrics

To evaluate the effectiveness of the proposed framework, we employ accuracy (Acc) and Area Under the Curve (AUC) as evaluation metrics. The Acc is calculated following Mallen et al. [18], which assesses the precision of the generated answer $\mathcal{A}$ by verifying if any of its substrings exactly match the gold answers associated with the Q. The AUC quantifies the area under the accuracy curve plotted against K as Eq. (2), which is to comprehensively evaluate the performance across various K of the retrieved D.

$$\begin{aligned} AUC &= \int_{x_s}^{x_e} Acc(x)\,dx \\ &\approx \frac{1}{2}\sum_{i=1}^{x_e-x_s+1}(x_i - x_{i-1})\left[Acc(x_i) + Acc(x_{i-1})\right], \end{aligned} \tag{2}$$

where $K \in [x_s, x_e]$, and x_s and x_e denote the minimum and maximum number of supporting documents, respectively. A higher AUC value indicates superior overall performance across the spectrum of document quantities. Given that our proposed framework is specifically designed to enhance RAG through heterogeneous context fusion, we strategically partition the evaluation of AUC into two distinct intervals: normal interval ($I_n = [1, 10], K \in I_n$) and longer interval ($I_l = [6, 10], K \in I_l$). This division highlights the effectiveness of the concept-based heterogeneous context fusion framework, especially with more source documents, showing its robustness and adaptability under increasing heterogeneity.

5. Results and analysis

5.1. Performance evaluation

The evaluation results for the PopQA and EntityQuestions datasets are depicted in Figs. 3 and 4, providing graphical trends of Acc as K increases. Furthermore, Tables 3 and 4 present quantitative results of AUC for the datasets. These tables include the calculation of Δ, quantifying the improvement achieved by our proposed method over the Vanilla methods. Specifically, Δ is computed as $\Delta = AUC_{ours} - AUC_{Vanilla}$. The detailed evaluation results in Acc are provided in Tables A.2 and A.3.

A key finding reflected by Figs. 3 and 4 is the superior performance of the proposed method in scenarios with a large number of source documents. As K increases, especially within the longer context setting (I_l), the Acc of our method consistently outperforms that of various backbone LLMs coupled with other compression methods. This trend suggests that the concepts fused by our method effectively reduce interference from heterogeneous noise, enabling the LLMs to focus on essential knowledge. Moreover, the positive Δ values in Tables 3 and 4 for the I_l interval further highlight our framework's improvement over baseline methods when more source documents are involved. This demonstrates the effectiveness of the AMR-driven distillation algorithm in extracting informative concepts from heterogeneous contexts, enabling LLMs to generate accurate answers even when confronted with noisy multi-source inputs. When using the Bloom-560m model as the backbone LLM, Δ exhibits negative trends in the I_n interval of both datasets, and SelCon does not perform well either. We hypothesize that small-scale models have a limited capacity to associate semantic scenarios through discrete concepts. This limitation hinders their ability to understand the core information expressed in the concept-fused context. Conversely, when coupling advanced LLMs such as LLaMA-2, the contexts produced by the proposed method and SelCon exhibit the most significant and second-most-significant improvements, respectively. This observation likely arises from these large-scale models' superior contextual understanding capabilities, which supports our hypothesis.

The proposed AMR-driven distillation method demonstrates clear advantages over prompt-based compression methods such as keyword extraction and summarization in fusing heterogeneous context. While these methods use LLMs to generate compressed representations and show competitive results in certain cases, they may inadvertently introduce noise or lose essential details during the compression process.

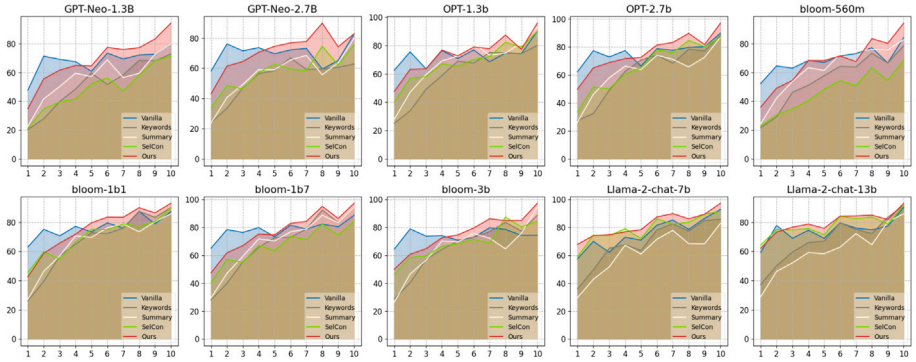


Fig. 3. The evaluation results of the $Acc \uparrow$ trends and $AUC \uparrow$ on the PopQA dataset. The vertical axis represents Acc, and the horizontal axis represents the number of supporting documents, K. The polyline reflects the changing trend of Acc with different K, and the under area is AUC.

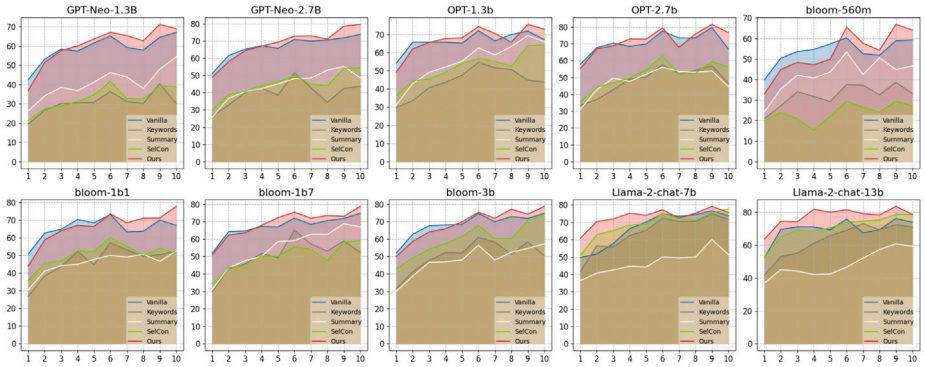


Fig. 4. The evaluation results of the $Acc \uparrow$ trends and $AUC \uparrow$ on the EntityQuestions dataset. The definitions of the axis and symbols are the same in Fig. 3.

Table 3
The quantitative results of $AUC \uparrow$ for the PopQA dataset, where the full name order of the LLMs is: GPT-Neo-1.3B, GPT-Neo-2.7B, OPT-1.3b, OPT-2.7b, Bloom-560m, Bloom-1b1, Bloom-1b7, Bloom-3b, LLaMA-2-chat-7b, LLaMA-2-chat-13b. The best results are in **bold**, and the second best results are in underlined. The increased and decreased Δ are marked differently.

D	K	G-1.3	G-2.7	O-1.3	O-2.7	b-560	b-1b1	b-1b7	b-3	L-7	L-13
Vanilla	I_n	620.68	631.39	656.68	687.15	**619.86**	**692.68**	**707.25**	671.88	682.30	672.03
	I_l	291.08	275.32	300.85	322.23	294.94	325.37	326.29	305.91	337.19	312.62
Keywords	I_n	468.94	484.98	554.67	571.38	502.70	610.69	621.85	600.65	628.78	617.06
	I_l	257.12	244.24	297.70	305.64	275.39	327.70	338.01	318.37	326.41	315.93
Summary	I_n	517.57	513.37	619.78	575.32	573.95	608.41	637.55	591.12	564.51	553.24
	I_l	263.14	260.64	316.80	290.50	304.55	313.36	336.20	297.44	291.50	291.39
SelCon	I_n	444.29	524.54	615.78	607.12	423.22	634.81	606.15	625.66	715.90	703.29
	I_l	237.49	262.78	313.39	323.69	230.20	318.64	306.72	314.07	344.10	332.51
Ours	I_n	**625.31**	**652.71**	**668.86**	**688.47**	608.31	686.29	698.91	**681.22**	**738.82**	**716.55**
	I_l	**322.37**	**321.73**	**329.65**	**344.31**	**314.34**	**347.71**	**355.52**	**344.08**	**357.56**	**339.38**
Δ	I_n	+4.63	+21.32	+12.18	+1.32	−11.55	−6.93	−8.34	+9.34	+56.52	+44.52
	I_l	+31.29	+46.41	+28.8	+22.08	+19.40	+22.34	+29.23	+38.17	+20.37	+26.76

Table 4
The quantitative results of $AUC \uparrow$ for the EntityQuestions dataset. The LLMs' order and symbol definitions are the same as Table 3.

D	K	G-1.3	G-2.7	O-1.3	O-2.7	b-560	b-1b1	b-1b7	b-3	L-7	L-13
Vanilla	I_n	531.54	605.06	602.52	634.28	**488.95**	594.88	608.85	619.30	607.22	632.24
	I_l	247.50	284.47	277.47	299.03	222.99	266.91	284.00	289.26	289.95	287.48
Keywords	I_n	280.76	360.00	403.37	439.73	295.02	428.54	465.15	462.65	584.67	574.61
	I_l	134.96	167.13	196.04	215.41	143.68	207.59	227.84	223.38	287.84	284.53
Summary	I_n	366.73	406.72	501.51	446.50	388.36	415.61	501.90	435.49	425.70	438.31
	I_l	179.97	205.02	255.51	210.93	187.75	197.43	257.16	211.83	210.34	222.92
SelCon	I_n	298.49	405.22	471.36	468.18	215.52	460.37	451.41	539.49	623.91	641.01
	I_l	144.69	195.05	231.76	223.55	108.45	214.94	217.40	261.79	295.33	304.57
Ours	I_n	**551.50**	**618.18**	**609.88**	**652.48**	483.02	**600.72**	**624.53**	**621.36**	**664.18**	**703.67**
	I_l	**267.12**	**298.74**	**285.06**	**303.49**	**243.55**	**286.20**	**295.45**	**300.29**	**303.39**	**320.87**
Δ	I_n	+19.96	+13.12	+7.36	+18.2	−5.93	+5.84	+15.58	+2.06	+56.96	+71.43
	I_l	+19.62	+14.27	+7.59	+4.45	+20.56	+19.29	+11.45	+11.03	+13.44	+33.39

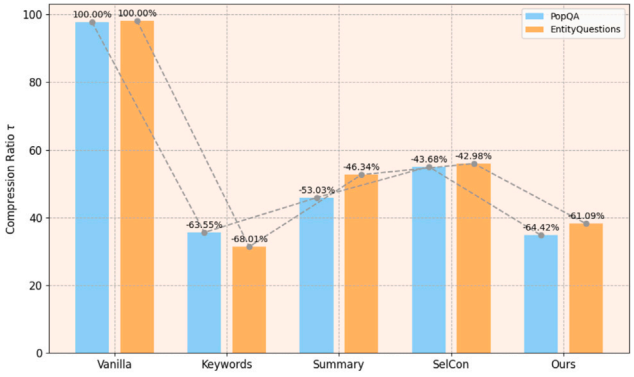


Fig. 5. The token-level compression ratio of different context compression methods. Each bar shows the compression ratio τ, the token length retained relative to the Vanilla method (100.00%). For the compression methods, the percentage above each bar reports the corresponding reduction ratio, $1 - \tau$.

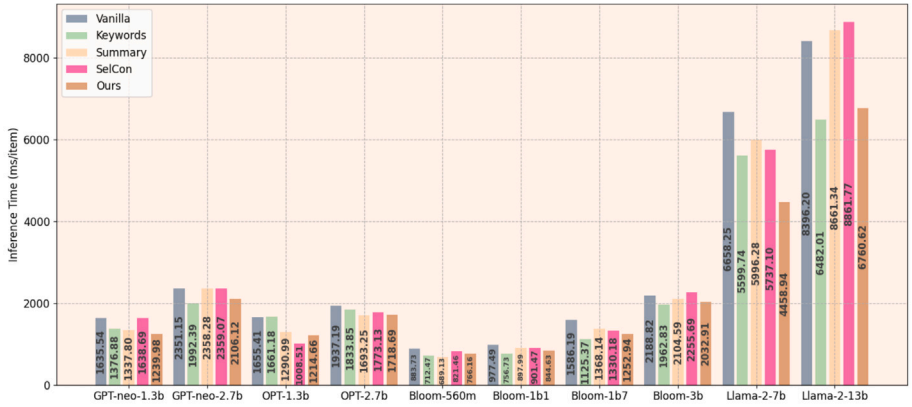


Fig. 6. The inference time comparison of different context compression methods. The values in the histogram represent the inference time for each model on a single item, measured in milliseconds.

Moreover, the generative nature of these methods makes them inherently difficult to control, even when provided with instructions as constraints. As a result, the generated keywords and summaries may exhibit randomness, potentially deviating from the core concepts conveyed in the original supporting documents. In contrast, our framework leverages the inherent structured semantic representation of AMR to capture the core concepts explicitly. This concept-based fusion approach enables the framework to faithfully align and integrate heterogeneous contextual representations, providing more reliable support for the RAG inference process than the multi-round inference-based manner.

SelCon is a linguistic context compression baseline. It identifies and prunes redundant tokens based on self-information computed at the lexical level. Compared with SelCon, the proposed concept-based fusion method operates at the semantic level and demonstrates superior performance. The effectiveness of SelCon depends on accurately determining the granularity for redundancy removal, which makes it sensitive to the choice of lexical units. In contrast, our method takes a higher-level view by focusing on the semantic consistency carried by the AMR structure, thereby reducing sensitivity to lexical biases and document-specific noise.

This insensitivity to lexical variability allows our method to function as a robust, plug-and-play heterogeneous context fusion component within various RAG systems. The advantage is most evident when the contexts contain irrelevant information and potential lexical errors. The robustness of the proposed framework is further validated by its consistent performance enhancements across a range of LLMs. These findings suggest that the concept-based heterogeneous context fusion framework can be effectively combined with diverse LLMs, providing a broadly applicable solution for enhancing RAG in knowledge-intensive scenarios.

5.2. Efficiency evaluation

5.2.1. Compression ratio

This section compares the token-level compression ratio achieved by different context fusion and compression methods. Fig. 5 presents the compression ratio (τ) following the calculation of $\tau = \tilde{L}/L$, where $\tilde{L}$ and L represent the token-level length of the fused or compressed context and the Vanilla context.

The findings demonstrate that the proposed AMR-driven concept distillation algorithm achieves competitive context-compression performance. Compared to the standard Vanilla method, our method can compress the average token-level length by over 60% while preserving the core concepts to the greatest extent. Notably, our method achieves the highest compression ratio on the PopQA dataset. In contrast, although the Keywords-based compression method exhibits a better compression ratio on the EntityQuestions dataset compared to ours, its performance lags behind the proposed algorithm in preserving core concepts. Compared to the other two baselines, our method shows a significant advantage in compression ratio.

5.2.2. Inference latency

This section analyses the inference latency by presenting the inference time comparison of backbone LLMs with different fusion and compression methods. The result is shown in Fig. 6. The inference latency evaluated here is based on the NVIDIA RTX A5000 GPU. The findings reveal that the context fused by our method demonstrates

advantages in the inference performance of most LLMs. Moreover, the context compressed by the Keywords method also exhibits shorter inference times, which is consistent with the trend observed in the previous compression ratio analysis.

6. Conclusions and future research

This paper presents a concept-based heterogeneous context fusion framework driven by AMR distillation. The framework uses AMR graphs to distil essential concepts from multi-source retrieved documents into a compact, unified set, which allows LLMs to focus on the most supportive knowledge for accurate question answering. By operating on the semantic structure exposed by AMR, the distillation algorithm normalizes surface-level variations across documents and produces cross-document consistent concepts with reduced noise. Because it does not modify the backbone model, the method works as a plug-and-play component across diverse backbone LLMs and can be attached to existing RAG systems to shorten the retrieved context and reduce inference cost. To the best of our knowledge, this is the first work to introduce AMR-driven semantic distillation for multi-source heterogeneous context fusion.

The framework still leaves room for improvement in a few aspects. The quality of the distilled concepts depends in part on the underlying AMR parser, and the benefit tends to be smaller for small-scale backbone models, whose more limited capacity makes it harder to associate semantic scenarios from discrete concepts. Moreover, we will study the sensitivity of the algorithm to character-level length and identify the conditions under which concept distillation gives the largest gains. We also plan to extend the framework to additional structured modalities, such as tabular data and knowledge base entries, and to evaluate it on a wider range of knowledge-intensive tasks and larger instruction-tuned models. These studies will clarify the applicability of the method and support its use across knowledge-intensive scenarios.

CRediT authorship contribution statement

Kaize Shi: Writing – original draft, Methodology, Conceptualization. **Xueyao Sun:** Software, Data curation. **Qika Lin:** Validation, Investigation. **Yan Li:** Writing – review & editing. **Qing Li:** Writing – review & editing, Supervision. **Guandong Xu:** Writing – review & editing, Supervision, Resources, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. AMR examples

This appendix presents an example of an AMR graph parsed from the supporting document, "`Alexander Rinnooy Kan of Amsterdam. In 1972-73, he worked as a mathematician at Spectrum Encyclopedia`". Fig. A.1 showcases the `DFS(·)` order for traversing the nodes in the AMR graph. Table A.1 shows the parsed AMR graph marked with the distilled concepts, demonstrating that the `DFS(·)` traversal maintains the relative position of adjacent concepts contained in the raw retrieved supporting document. This example illustrates how the AMR-driven distillation algorithm captures key concepts to produce a unified concept set.

Table A.1
Example of the AMR-based concept distillation. The nodes carrying concepts are **bolded**, including "Alexander Rinnooy Kan, Amsterdam. work, mathematics, Spectrum Encyclopedia, 1972, 1973". The special roles, `:wiki`, `:year`, and `:name` are highlighted.

Supporting Document: Alexander Rinnooy Kan of Amsterdam. In 1972–73, he worked as a mathematician at Spectrum Encyclopedia.

```
(m / multi-sentence
    :snt1 (p / person
            :name (n / name
                    :op1 "Alexander"
                    :op2 "Rinnooy"
                    :op3 "Kan")
            :location (c / city
                    :wiki "Amsterdam"
                    :name (n2 / name
                            :op1 "Amsterdam")))
    :snt2 (w / work-01
            :ARG0 (h / he)
            :ARG1 (m2 / mathematics)
            :ARG2 (r / research-institute
                    :wiki "Spectrum_Encyclopedia"
                    :name (n3 / name
                            :op1 "Spectrum"
                            :op2 "Encyclopedia"))
            :time (d / date-interval
                    :op1 (d2 / date-entity
                            :year 1972)
                    :op2 (d3 / date-entity
                            :year 1973))))
```

Appendix B. Accuracy details

The detailed *Acc* for the different context fusion and compression methods on PopQA and EntityQuestions datasets are presented in Tables A.2 and A.3, respectively.

Appendix C. Random concept traversal

A significant challenge in long context-based LLMs inference is the "lost-in-the-middle" phenomenon [7], where LLMs tend to overlook key features that appear in the middle of the context. Therefore, it is important to verify the robustness of the fused context to changes in the positional arrangement of key information. We employ random traversal strategies `RT(·)`, including global random traversal `GRT(·)` and local random traversal `LRT(·)`, to empirically study whether heterogeneous context fusion can mitigate the impact of this phenomenon on LLMs' inference.

C.1. Global random traversal

Global random traversal `GRT(·)` involves randomly traversing concept nodes throughout the AMR graph, covering all sentences in the supporting documents. Concepts obtained in this setting have no specific positional correlation. The results are presented in Tables A.4 and A.5. These results reveal changes in *Acc* due to the concepts derived from global random traversal. Apart from a few cases where *Acc* is higher than the concepts from the AMR-driven concept distillation algorithm, most of the other *Acc* values corresponding to the same settings decrease.

C.2. Local random traversal

Local random traversal `LRT(·)` involves randomly traversing concept nodes within the sub-AMR graph corresponding to a specific sentence. In this setting, the distilled concepts do not retain positional correlations within individual sentences but do preserve them across sentences in the retrieved document. The results are presented in Tables A.6 and A.7. The accuracy (*Acc*) also exhibits a declining trend with a minor decline.

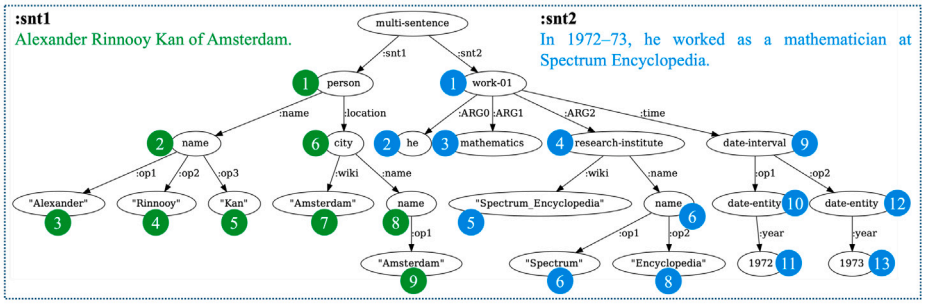


Fig. A.1. The node order of the DFS(·) traversal within the AMR graph, with nodes marked in different colors for the two sentences. The marked nodes here are not the final distilled concepts. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table A.2
Accuracy (*Acc* ↑) comparison on the PopQA dataset. The best results for each LLM with setting K are in **bold**, and the second best results are in underlined. Δ here represents the difference between the methods of "Ours" and "Vanilla", and the increased and decreased Δ are marked differently. The best results for each of K are marked.

LLMs	C \ K	1	2	3	4	5	6	7	8	9	10
GPT-Neo-1.3B	Vanilla	**47.69**	**71.38**	**69.19**	**67.56**	60.87	73.51	69.44	72.15	72.73	80.00
	Keywords	20.46	27.85	39.81	48.47	59.63	51.66	58.33	68.35	68.18	72.86
	Summary	22.76	41.70	50.24	59.54	57.14	68.87	56.48	59.49	72.73	80.00
	SelCon	21.14	34.74	39.57	41.60	52.17	56.29	47.22	58.23	68.18	71.43
	Ours	34.96	55.62	61.61	64.89	**64.60**	**77.48**	**75.93**	**77.22**	**83.33**	**94.29**
	Δ	−12.73	−15.76	−7.58	−2.67	+3.73	+3.97	+6.49	+5.07	+10.60	+14.29
GPT-Neo-2.7B	Vanilla	**58.13**	**76.13**	**71.56**	**73.66**	69.56	72.19	73.15	59.49	65.15	**82.86**
	Keywords	23.98	33.51	46.92	56.49	58.39	66.89	59.26	59.49	60.61	62.86
	Summary	23.71	40.63	49.29	59.16	59.01	65.56	68.52	55.70	63.64	80.00
	SelCon	33.88	48.58	46.45	57.25	62.73	59.60	58.33	74.68	62.12	75.71
	Ours	43.09	61.44	64.45	70.61	**74.53**	**76.82**	**77.78**	**89.87**	**74.24**	**82.86**
	Δ	−15.04	−14.69	−7.11	−3.05	+4.97	+4.63	+4.63	+30.38	+9.09	0.00
OPT-1.3b	Vanilla	**62.47**	**75.52**	**63.51**	76.34	70.81	76.82	68.52	74.68	74.24	90.00
	Keywords	25.07	33.89	49.05	58.78	68.94	67.55	75.00	74.68	74.24	80.00
	Summary	27.91	47.05	59.48	69.08	**72.67**	**81.46**	73.15	74.68	**81.82**	92.86
	SelCon	40.11	56.77	58.06	67.18	65.22	70.20	72.22	82.28	78.79	90.00
	Ours	47.56	63.12	**63.51**	**76.72**	**72.67**	78.81	**77.78**	**87.34**	77.27	**95.71**
	Δ	−14.91	−12.40	0.00	+0.38	+1.86	+1.99	+9.26	+12.66	+3.03	+5.71
OPT-2.7b	Vanilla	**62.33**	**77.43**	**72.99**	**77.48**	66.45	78.80	77.78	79.75	80.30	90.00
	Keywords	27.51	32.44	49.05	62.60	70.81	74.17	68.52	78.48	77.27	88.57
	Summary	26.29	46.14	58.06	66.41	63.98	74.17	71.30	65.82	72.72	87.14
	SelCon	33.74	51.34	50.24	60.69	65.22	78.15	75.93	84.81	80.30	87.14
	Ours	49.86	65.11	68.96	71.76	**72.67**	**81.46**	**83.33**	**89.87**	**81.81**	97.14
	Δ	−12.47	−12.32	−4.03	−5.72	+6.22	+2.66	+5.55	+10.12	+1.51	+7.14
Bloom-560m	Vanilla	**52.30**	**64.58**	**63.03**	**68.32**	67.08	71.52	73.14	77.22	66.67	84.29
	Keywords	21.68	29.23	46.21	51.15	57.76	64.24	63.89	73.42	66.67	78.57
	Summary	24.39	42.16	53.79	63.36	61.49	**72.82**	**75.00**	75.95	75.76	82.86
	SelCon	22.90	30.68	34.83	40.46	48.45	54.30	50.93	63.29	54.55	68.57
	Ours	35.91	49.12	54.50	**68.32**	**68.32**	71.52	67.59	**83.54**	**80.30**	**94.29**
	Δ	−16.39	−15.46	−8.53	0.00	+1.24	0.00	−5.55	+6.32	+13.63	+10.00
Bloom-1b1	Vanilla	**63.01**	**75.06**	**70.62**	**77.10**	73.29	79.47	75.93	87.34	78.79	87.14
	Keywords	26.15	39.40	56.87	64.89	72.67	72.19	75.93	87.34	83.33	90.00
	Summary	27.10	46.37	56.87	70.61	69.57	76.16	78.70	73.42	80.30	85.71
	SelCon	45.53	59.68	54.74	66.41	75.16	74.83	78.70	77.22	80.30	90.00
	Ours	42.55	58.68	65.64	71.76	79.50	**83.44**	**83.33**	**89.87**	**86.36**	**92.86**
	Δ	−20.46	−16.38	−4.98	−5.34	+6.21	+3.97	+7.40	+2.53	+7.57	+5.72
Bloom-1b7	Vanilla	**65.04**	**78.35**	**76.30**	79.77	73.29	81.46	78.70	82.28	80.30	88.57
	Keywords	28.05	39.86	55.21	65.65	72.67	72.85	78.70	92.41	83.33	94.29
	Summary	29.54	46.98	59.24	71.76	70.19	76.82	78.70	88.61	83.33	94.29
	SelCon	40.51	57.15	54.74	67.18	63.35	73.51	71.30	82.28	74.24	84.29
	Ours	47.43	61.74	66.82	75.19	**74.53**	**82.78**	**84.26**	94.94	**86.36**	97.14
	Δ	−17.61	−16.61	−9.48	−4.58	+1.24	+1.32	+5.56	+12.66	+6.06	+8.57
Bloom-3b	Vanilla	**64.36**	78.81	**73.70**	**74.05**	70.80	72.85	79.62	78.48	74.24	74.29
	Keywords	28.86	40.24	56.87	64.12	70.19	72.85	76.85	83.54	77.27	88.57
	Summary	26.29	46.60	56.40	70.22	69.57	75.50	72.22	64.56	75.76	94.29
	SelCon	46.34	58.45	59.48	66.41	68.32	71.52	68.52	**87.34**	80.30	84.29
	Ours	50.00	60.67	64.69	72.52	**74.53**	**79.47**	**86.11**	84.81	**84.85**	97.14
	Δ	−14.36	−18.14	−9.01	−1.53	+3.73	+6.62	+6.49	+6.33	+10.61	+22.85
LLaMA-2-c-7	Vanilla	57.32	69.93	62.09	72.90	70.81	81.45	85.19	78.48	86.36	92.86
	Keywords	35.91	49.50	64.93	67.56	63.35	78.15	82.41	77.22	84.85	85.71
	Summary	29.54	42.77	51.66	67.18	60.87	71.52	77.78	68.35	68.18	82.86
	SelCon	58.81	**74.37**	73.93	**79.00**	72.05	86.09	82.41	83.54	89.39	91.43
	Ours	67.74	74.06	**74.64**	76.72	**78.26**	87.42	89.81	**86.08**	89.39	97.14
	Δ	+10.42	+4.13	+12.55	+3.82	+7.45	+5.97	+4.62	+7.60	+3.03	+4.28
LLaMA-2-c-13	Vanilla	59.21	**77.74**	68.96	74.43	68.94	79.47	75.93	74.68	77.27	90.00
	Keywords	37.40	50.11	59.48	66.03	67.08	79.47	75.00	72.15	**83.33**	91.43
	Summary	28.86	46.29	52.13	59.16	58.39	62.91	72.22	64.56	80.30	85.71
	SelCon	**64.36**	74.90	74.64	75.57	71.43	**84.11**	82.41	83.54	78.79	91.43
	Ours	62.06	73.14	76.54	**78.63**	**75.78**	**84.11**	**84.26**	**84.81**	81.82	**92.86**
	Δ	+2.85	−4.60	+7.58	+4.20	+6.84	+4.64	+8.33	+10.13	+4.55	+2.86

Table A.3
Accuracy (*Acc* ↑) comparison on EntityQuestions dataset. The symbols definitions are as Table A.2.

LLMs	C \ K	1	2	3	4	5	6	7	8	9	10
GPT-Neo-1.3B	Vanilla	**42.43**	**53.15**	**58.34**	57.27	61.49	65.15	59.18	57.83	64.42	66.99
	Keywords	19.39	26.62	30.15	30.40	30.75	36.36	31.12	30.12	40.49	30.10
	Summary	26.21	34.07	38.51	36.78	41.19	46.21	43.88	37.95	47.85	54.36
	SelCon	21.13	27.68	28.81	31.28	34.63	41.67	32.65	32.53	39.26	38.83
	Ours	36.74	51.38	57.61	**59.91**	**63.58**	**67.05**	**65.31**	**62.65**	**71.17**	**68.93**
	Δ	−5.69	−1.77	−0.73	+2.64	+2.09	+1.90	+6.13	+4.82	+6.75	+1.94
GPT-Neo-2.7B	Vanilla	**51.05**	**61.58**	**65.22**	**67.18**	65.67	70.83	69.90	70.48	71.78	73.79
	Keywords	26.70	32.74	40.00	42.51	38.51	51.51	42.86	34.34	42.33	43.69
	Summary	25.91	36.73	40.60	42.29	45.07	48.11	48.47	53.01	55.21	48.54
	SelCon	30.04	38.86	40.75	44.27	46.27	50.00	44.90	43.98	53.99	54.36
	Ours	48.95	58.03	64.33	67.00	**69.25**	**72.72**	**72.96**	**71.08**	**78.53**	79.61
	Δ	−2.10	−3.55	−0.89	−0.18	+3.58	+1.89	+3.06	+0.60	+6.75	+5.82
OPT-1.3b	Vanilla	**54.04**	**65.66**	**65.67**	65.64	65.07	71.97	66.33	**69.88**	71.78	66.99
	Keywords	29.68	33.54	40.60	43.61	47.46	54.55	51.53	50.60	44.79	43.69
	Summary	31.18	42.86	49.10	51.98	55.22	62.50	58.67	63.25	69.33	66.02
	SelCon	36.27	44.54	45.37	49.12	54.03	56.82	55.10	52.41	63.80	64.08
	Ours	49.13	61.93	65.52	**67.62**	**68.06**	**74.24**	**70.41**	65.66	**75.46**	**72.82**
	Δ	−4.91	−3.73	−0.15	+1.98	+2.99	+2.27	+4.08	−4.22	+3.68	+5.83
OPT-2.7b	Vanilla	**57.87**	**67.70**	**70.44**	68.50	69.85	77.65	**73.47**	73.49	79.75	66.99
	Keywords	33.51	36.82	42.69	49.12	50.15	57.58	52.04	54.22	57.06	46.60
	Summary	31.11	42.77	49.70	47.58	51.94	56.06	53.57	53.01	53.99	44.66
	SelCon	35.43	44.19	46.87	49.78	54.63	62.88	52.04	52.41	59.51	56.31
	Ours	55.36	66.99	68.81	**72.91**	**72.83**	**79.55**	67.86	**75.90**	**81.60**	**76.70**
	Δ	−2.51	−0.71	−1.63	+4.41	+2.98	+1.90	−5.61	+2.41	+1.85	+9.71
Bloom-560m	Vanilla	**39.86**	**50.40**	**53.58**	**54.63**	**57.31**	60.23	52.55	51.81	58.90	59.22
	Keywords	20.89	27.15	34.03	31.72	29.25	37.50	37.24	32.53	38.65	33.01
	Summary	24.18	35.40	42.09	40.75	43.58	53.41	42.35	50.60	44.79	46.60
	SelCon	19.87	23.96	20.90	15.42	22.09	29.55	26.53	24.10	29.45	27.18
	Ours	32.91	44.90	48.36	47.14	49.85	**65.53**	**57.65**	**54.22**	**66.87**	**64.08**
	Δ	−6.95	−5.50	−5.22	−7.49	−7.46	+5.30	+5.10	+2.41	+7.97	+4.86
Bloom-1b1	Vanilla	**50.75**	**62.64**	**64.78**	**70.26**	**68.36**	73.11	63.27	63.65	69.94	66.99
	Keywords	26.81	38.24	43.73	52.20	44.78	57.20	53.06	49.40	50.31	52.43
	Summary	30.46	41.08	44.18	44.93	47.76	50.00	48.98	50.60	46.63	52.43
	SelCon	35.73	45.70	46.87	52.64	52.24	60.23	55.10	50.00	53.99	51.46
	Ours	43.87	58.74	63.88	66.96	66.27	**73.48**	**68.37**	**71.08**	**71.17**	**77.67**
	Δ	−6.88	−3.90	−0.90	−3.30	−2.09	+0.37	+5.10	+7.43	+1.23	+10.68
Bloom-1b7	Vanilla	**51.89**	**64.24**	**64.63**	67.18	66.87	71.97	68.37	70.48	71.78	74.76
	Keywords	29.74	43.03	45.82	51.76	49.25	65.15	57.14	53.01	58.90	52.43
	Summary	31.18	43.48	47.61	50.00	58.51	59.09	62.76	62.65	68.71	66.99
	SelCon	34.41	43.66	44.93	50.22	50.15	55.68	54.08	47.59	58.28	59.22
	Ours	50.87	62.29	63.58	**67.84**	**72.24**	**75.38**	**71.94**	**73.49**	**73.01**	**78.64**
	Δ	−1.02	−1.95	−1.05	+0.66	+5.37	+3.41	+3.57	+3.01	+1.23	+3.88
Bloom-3b	Vanilla	**51.94**	**62.73**	**67.61**	**68.06**	68.36	74.62	69.90	72.89	71.78	74.76
	Keywords	31.60	41.08	47.76	52.20	51.94	60.98	58.16	51.20	58.28	50.49
	Summary	30.10	38.69	46.72	46.92	48.06	56.44	47.96	52.41	54.60	57.28
	SelCon	42.97	49.16	54.18	57.49	61.49	67.80	60.20	60.24	70.55	73.79
	Ours	49.97	58.65	64.03	65.86	**69.85**	**75.38**	**71.94**	**77.11**	**74.23**	**78.64**
	Δ	−1.97	−4.08	−3.58	−2.20	+1.49	+0.76	+2.04	+4.22	+2.45	+3.88
LLaMA-2-c-7	Vanilla	49.67	51.73	57.76	66.30	70.15	75.00	**73.47**	74.39	76.69	73.79
	Keywords	41.53	56.52	55.67	62.33	65.37	72.35	70.41	70.48	74.85	71.84
	Summary	36.39	40.73	42.54	44.71	44.18	50.00	49.49	50.00	60.12	51.46
	SelCon	50.99	62.91	65.22	68.50	68.96	75.00	72.45	71.08	75.46	**77.67**
	Ours	**60.80**	**70.45**	**71.94**	**75.33**	**74.03**	**77.27**	72.45	**75.30**	**79.14**	75.73
	Δ	+11.13	+18.72	+14.18	+9.03	+3.88	+2.27	−1.02	+0.91	+2.45	+1.94
LLaMA-2-c-13	Vanilla	52.36	69.48	71.04	70.93	69.25	75.76	67.35	69.28	76.07	73.79
	Keywords	41.77	52.97	55.07	61.01	65.67	68.94	72.96	69.28	72.39	70.87
	Summary	36.80	45.06	44.18	42.07	42.39	46.59	52.04	57.23	60.74	59.22
	SelCon	52.54	65.04	69.25	68.50	70.45	73.86	74.49	75.30	78.53	**78.64**
	Ours	63.61	74.37	74.18	81.72	80.00	81.44	79.08	78.31	83.44	**78.64**
	Δ	+11.25	+4.89	+3.14	+10.79	+10.75	+5.68	+11.73	+9.03	+7.37	+4.85

Table A.4
The Acc ↑ results on the PopQA dataset by concept obtained with the global random traverse. Any changes compared to the original proposed AMR-based concept distillation algorithm are marked.

LLMs \ K	1	2	3	4	5	6	7	8	9	10
G-1.3	34.96	55.62	60.43	64.50	63.35	76.82	75.00	75.93	80.30	94.29
G-2.7	43.09	61.36	63.51	69.85	74.53	72.84	77.78	89.87	74.24	82.86
O-1.3	47.56	63.12	63.51	76.34	72.05	74.17	77.78	87.34	77.27	95.71
O-2.7	49.86	65.11	68.96	71.76	70.80	80.79	81.48	89.87	81.81	90.00
b-560	35.91	48.58	53.08	67.56	68.32	71.52	67.59	83.54	80.30	94.29
b-1b1	42.28	58.68	65.40	71.76	78.81	83.44	83.33	88.61	86.36	90.00
b-1b7	47.43	61.51	66.82	74.81	73.91	82.78	84.26	93.67	84.85	95.71
b-3	49.73	60.67	64.22	71.76	74.53	79.47	85.19	84.81	83.33	95.71
L-7	62.73	74.06	74.64	76.72	78.26	87.42	85.19	86.08	86.36	92.86
L-13	62.06	73.14	76.54	78.63	75.78	84.11	82.41	83.54	78.79	90.00

Table A.5
The Acc ↑ results on the EntityQuestions dataset by concept obtained with the global random traverse. The symbol definitions are the same with Table A.4.

LLMs \ K	1	2	3	4	5	6	7	8	9	10
G-1.3	36.74	51.38	57.61	59.91	61.49	67.05	65.31	57.83	71.17	66.99
G-2.7	48.95	58.03	64.33	66.96	65.67	70.83	72.96	70.48	78.53	73.79
O-1.3	49.13	61.93	65.52	67.62	65.07	71.97	70.41	65.66	75.46	66.99
O-2.7	55.36	66.99	68.81	72.91	69.85	77.65	67.86	73.49	81.60	66.99
b-560	32.91	44.90	48.36	47.14	49.85	60.23	57.65	51.81	66.87	59.22
b-1b1	43.87	58.74	63.88	66.96	66.27	73.11	68.37	62.65	71.17	66.99
b-1b7	50.87	62.29	63.58	67.84	66.87	75.38	68.37	73.49	71.78	74.76
b-3	49.97	58.65	64.03	65.86	68.36	74.62	69.90	72.89	74.23	74.76
L-7	60.80	70.45	71.94	75.33	74.03	77.27	72.45	75.30	79.14	73.79
L-13	63.61	74.37	74.18	81.72	80.00	81.44	79.08	78.31	83.44	78.64

Table A.6
The Acc ↑ results on the PopQA dataset by concept obtained with the local random traverse. Any changes compared to the original proposed AMR-based concept distillation algorithm are marked.

LLMs \ K	1	2	3	4	5	6	7	8	9	10
G-1.3	34.55	55.39	60.90	64.89	63.98	76.82	75.93	75.95	80.30	94.29
G-2.7	42.82	61.36	63.51	69.85	73.29	72.85	77.78	89.87	72.73	81.43
O-1.3	47.56	62.82	63.03	75.57	72.67	75.50	76.85	87.34	77.27	95.71
O-2.7	49.86	65.11	68.96	71.37	70.81	80.79	81.48	89.87	81.81	90.00
b-560	35.37	48.43	54.98	67.56	68.32	71.52	65.79	83.54	80.30	92.86
b-1b1	42.55	58.45	65.88	71.37	79.50	83.44	83.33	87.34	86.36	90.00
b-1b7	47.56	61.44	66.82	75.19	73.91	82.78	83.33	93.67	84.85	94.29
b-3	49.59	60.67	64.21	71.76	74.53	79.47	86.11	84.81	83.33	94.29
L-7	62.60	74.06	74.64	76.72	78.88	87.42	83.33	86.08	84.85	91.43
L-13	61.52	73.60	76.30	77.86	75.16	84.11	83.33	83.54	75.76	91.43

Table A.7
The Acc ↑ results on the EntityQuestions dataset by concept obtained with the local random traverse. The symbol definitions are the same with Table A.6.

LLMs \ K	1	2	3	4	5	6	7	8	9	10
G-1.3	36.44	51.20	57.61	59.91	61.49	66.67	65.31	62.65	71.17	66.02
G-2.7	48.89	58.30	64.48	67.00	65.67	72.21	72.96	71.08	78.53	79.61
O-1.3	49.07	62.20	65.22	67.62	68.06	74.24	70.41	65.66	73.62	72.82
O-2.7	55.36	66.90	68.81	72.69	72.83	77.65	67.35	73.49	82.20	76.70
b-560	32.91	44.90	47.76	47.14	49.25	65.53	56.63	54.22	65.03	60.19
b-1b1	43.87	58.65	63.88	66.96	65.67	72.73	68.37	71.08	70.55	77.67
b-1b7	50.87	62.29	64.03	67.84	72.24	75.38	68.37	74.09	71.16	73.79
b-3	49.85	58.39	64.03	65.86	68.36	75.38	69.90	77.11	75.46	78.64
L-7	59.90	70.90	72.39	73.73	74.03	77.27	72.45	75.30	80.37	74.76
L-13	64.09	73.82	73.43	81.72	80.00	80.30	78.57	77.11	82.82	78.64

Data availability

The authors do not have permission to share data.

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